

USER MANUAL

1. Introduction

This user manual provides a step-by-step guide to using the Fairness-Aware AI Recruitment System. The system is designed to assist recruiters in making unbiased hiring decisions and allows candidates to apply for jobs and track their application status. It integrates machine learning, fairness evaluation, and explainability features to ensure transparency and efficiency in recruitment.

2. System Overview

The system has two roles:

- Recruiter
- Candidate

Each user interacts with the system through a web-based interface.

3. System Requirements

- Web browser
- Internet connection

4. User Roles and Functionalities

4.1 Recruiter

The recruiter can:

- Post job openings
- View candidate applications
- See shortlisted/rejected candidates
- View fairness-aware results
- Analyze model decisions using SHAP and LIME
- Access recruiter dashboard

4.2 Candidate

The candidate can:

- View available job postings
- Apply for jobs
- Receive selection/rejection notifications

- Track application status
- Access candidate dashboard

5. System Workflow

Step 1: Login/Register (In navigation bar you can view login and register buttons)

- User logs into the system using credentials
- Role-based access is provided (Recruiter/Candidate)

Step 2: Recruiter Actions

1. Navigate to Job Posting Page (In navigation bar you can see a button jobs)
2. Enter job details:
 - Role
 - Skills required
 - Experience
 - Location
 - Salary
3. Submit job

Job becomes visible to candidates

Step 3: Candidate Actions

1. View available jobs
2. Select a job
3. Click Apply
4. Submit application

Step 4: AI-Based Evaluation

- System processes candidate data
- Machine Learning model predicts:
 - Selected / Not Selected
- Fairness metrics are calculated:
 - SPD
 - DI
 - EOD

Step 5: Recruiter Analysis

- Recruiter views:
 - Candidate results
 - Fairness metrics
- SHAP & LIME explanations show:
 - Why a candidate was selected/rejected

Step 6: Notifications

- Candidate receives notification:
 - Shortlisted / Rejected

Step 7: Dashboard Access

Recruiter Dashboard:

- Hiring statistics
- Model insights
- Candidate results

Candidate Dashboard:

- Applied jobs
- Application status
- Notifications

6. Key Features

1. Job Management

Recruiters can create and manage job postings

2. AI-Based Candidate Screening

Automated candidate evaluation using ML models

3. Fairness Evaluation

Detects bias using:

Statistical Parity Difference

Disparate Impact

Equal Opportunity Difference

4. Bias Mitigation

Uses techniques like:

Reweighting

Adversarial Debiasing

5. Explainability

SHAP & LIME provide transparency in decisions

6. Dashboard Visualization

Displays results in graphs and summaries

7. Error Handling

- Invalid inputs are restricted

- Missing data prompts user warnings
- System ensures proper data validation

8. Security Features

- Role-based access control
- Secure login system
- Data handled securely in backend

9. Limitations

- Works on structured datasets
- Model performance depends on data quality
- Real-time updates may depend on system performance

10. Conclusion

The system provides a complete recruitment solution by combining automation, fairness, and explainability. It helps recruiters make unbiased decisions while providing candidates with a transparent application process.